

Molecular Drivers of Breast Cancer Metastasis: The Interplay Between EMT, Cell-Adhesion Pathways, and Hybrid Epithelial–Mesenchymal States

Zareen Suprova^{1,†}, Junaina Ariba Tiba^{1,†} and Oishika Sen^{1,†}

¹Ideal School and College

²Legend International School

³Govt. City College, Chattogram

[†]These authors contributed equally to this work.

Abstract

Breast cancer metastasis is driven by complex molecular processes involving **epithelial–mesenchymal transition (EMT)**, altered cell adhesion, and cellular plasticity. Increasing evidence suggests that metastatic cells may occupy **hybrid epithelial–mesenchymal (E/M) states** rather than undergoing a complete EMT. This study investigated molecular alterations in EMT- and cell-adhesion-associated genes and their relevance to epithelial–mesenchymal plasticity in breast cancer. A mixed-methods secondary research design was employed, integrating a structured literature review with genomic and mRNA expression data from publicly available cancer databases. Eight genes—**CDH1**, **CDH2**, **VIM**, **SNAI1**, **SNAI2**, **TWIST1**, **ZEB1**, and **ZEB2**—were analyzed using cBioPortal and UALCAN. **CDH1** showed the highest overall genomic alteration frequency (**11%**), while somatic mutation analysis identified a **13.1%** mutation frequency with multiple mutation types. mRNA analysis revealed heterogeneous expression patterns: CDH1 expression was relatively maintained in primary tumors, whereas CDH2 expression was increased. The remaining EMT-associated genes showed variable expression rather than a uniform mesenchymal pattern. The coexistence of CDH1 and increased CDH2 suggests the presence of molecular features compatible with **epithelial–mesenchymal plasticity** and provides basis for investigating potential **hybrid E/M states**. However, the bulk-tumor mRNA data do not establish hybrid EMT at the individual-cell level. Overall, the findings highlight heterogeneous molecular alterations in EMT and cell-adhesion pathways and provide a molecular framework for further investigation of hybrid EMT states in breast cancer metastasis.

Keywords: Breast cancer; Epithelial–mesenchymal transition (EMT); Hybrid epithelial–mesenchymal states; Epithelial–mesenchymal plasticity; E-cadherin; N-cadherin; Cell adhesion; Metastasis

Introduction

Breast cancer remains the most frequently diagnosed malignancy among women worldwide and continues to represent a significant cause of cancer-related mortality. Recent global estimates place the annual burden at over 2.3 million new cases and more than 660,000 deaths, with substantial regional disparities in outcomes tied to differences in early detection and treatment access [1]. Despite meaningful advances in screening, molecular diagnostics, and targeted therapy, metastatic disease remains responsible for the overwhelming majority of breast cancer deaths, underscoring the continued need to understand the biological mechanisms that allow tumor cells to escape the primary site and colonize distant organs. Metastasis is not a single event but a coordinated, multistep process. Malignant cells must detach from the primary tumor, invade the surrounding extracellular matrix, enter the circulatory or lymphatic system, survive the physical and immunological stresses of transit, exit into a new tissue, and adapt well enough

to that environment to establish a secondary growth. Modeling of this cascade suggests that survival within circulation is one of its most limiting steps, illustrating how inefficient dissemination is relative to the devastating outcomes it produces once successful [2]. Each transition in this sequence depends on the interaction of signaling pathways, adhesion molecules, and microenvironmental cues, meaning metastatic competence emerges from the cumulative effect of many molecular changes rather than any single alteration acting alone.

Among the molecular programs implicated in this cascade, epithelial–mesenchymal transition (EMT) has been widely recognized as a central driver of tumor cell invasion, migration, and therapeutic resistance. EMT is a reversible reprogramming process through which epithelial cells lose the features that anchor them within organized tissue and acquire a more motile, invasive phenotype [3]. This transition is orchestrated by a defined set of transcription factors—including SNAI1 (Snail), TWIST1, ZEB1, and ZEB2—that are activated by upstream signaling cues such as TGF- β and drive coordinated changes in gene expression, cytoskeletal organization, and intercellular adhesion [3], [4]. However, growing evidence indicates that many breast cancer